A MODERN LOOK AT CLASSICAL GENERALIZATION BOUNDS

MATI

Abstract.

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1. Introduction

In machine learning, there exists a large number of algorithms, notably neural networks trained with gradient descent, which achieve great performances at prediction tasks on unseen data. A big question in learning theory is to understand the relationship between the behavior of such algorithms on the available data and their future behavior on unseen data (generalization). How can we predict how this future behavior will look like?

A number of results have tried to answer this question [4, 2, 17, 8]. Recent works seem to suggest that the classical theory fails to explain generalization, in particular they fail to describe the behavior of the test error of learning algorithms as the comlpexity of the hypotheses set increases [3, 19, 20, 15].

This thesis seeks to understand whether this is true by testing theoretical results against empirical evidences. We study the setting of a finite hypotheses set of binary decision trees. We look for a bound on the test error as the maximum depth of the decision trees (i.e. the complexity of the hypotheses space). In particular, we want this function to qualitatively reproduce the test error curve as well as to be sensitive to changes in the structure of the learning problem that influence the generalization performance of the algorithm. We test a number of candidates coming from existing theoretical results against empirical evidence to analyze their validity.

Previous work

This section presents the state of the art and how the present work relates to it.

The behaviour of the test error of tree-based models as a function of their complexity has itself been the object of debate. [3] report a double descent curve for tree-based models, where the test error decreases again beyond the interpolation threshold. [6] show that this second descent arises from combining distinct complexity axes into a single one, such as the number of leaves of a tree and the number of trees in an ensemble, and that the curves recover a traditional convex shape when plotted against an effective number of parameters. In our setting, complexity is measured along a single axis, the maximum depth d of a single decision tree. Along this axis,

once the learned tree interpolates the training sample, a larger depth no longer changes the learned model, and the empirical test error saturates (Section 2); a second descent can only arise by moving to a different complexity axis, such as the number of trees in an ensemble.

Several works evaluate generalization bounds for decision trees in practice, but with a motivation different from ours. Specifically, [10] use Rademacher complexity as an empirical complexity measure: they compute the Rademacher penalty $\sup_{\ell_h \in \mathscr{L}} \left| \frac{1}{n} \sum_i \xi_i \ell_h(X_i, Y_i) \right|$ on a single realization of D and ξ, and plug it into a high-probability bound on $R(h)$. Their hypotheses set is data-dependent, namely the prunings of a tree grown on a separate sample, for which reduced error pruning applied to relabeled data computes the supremum exactly. They compare this bound with an Occam's razor bound based on a uniform code length over the prunings, analogous to our Theorem 2.6. Our work differs in that we consider hypotheses sets $\mathscr{H}_d$ fixed in advance and aim at interpretable analytical expressions rather than numerical values; hence, we use $\mathcal{R}_n(\mathscr{L})$ only as an intermediate step towards $N_D(\mathscr{H})$.

In [12], the authors derive an analytical upper bound on the growth function of classes of binary decision trees with fixed structure, that is, a structure-dependent count of the distinct classifications that such trees can realize on a sample, more detailed than the counting inequality $\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$ of Section 5. They use it within a structural risk minimization bound as a pruning criterion, evaluated through the test accuracy of the pruned trees. Our interest is different: rather than using a bound for model selection, we compare its shape with the empirical generalization gap as a function of the complexity of the hypotheses set.

Other works compare the numerical tightness of several generalization bounds on benchmark datasets [9, 16], or use bounds on the VC dimension of decision trees for pruning [18]. Further papers propose bounds on the error of decision trees that depend on the specific training sample or on an effective number of leaves [13, 7, 2], without evaluating them empirically. Outside decision trees, [8] test a large number of complexity measures against the generalization of neural networks.

Although not specific to decision trees, [4] is a survey on statistical bounds on the risk of learning algorithms, and has been a fundamental guide for the theoretical side of this thesis. A difference, however, lies in the emphasis of the analysis. The survey is largely concerned with how bounds behave as the sample size n grows, as guarantees on the precision of the empirical risk in estimating the true risk. The same perspective underlies the statistical analyses of CART [11]. We, instead, keep n fixed and regard bounds as functions of the complexity of the hypotheses set, which should describe the qualitative behaviour of the generalization gap. For instance, our Theorem 4.3, analogous to Theorem 5.1 of [4], is crucial to this work because it allows us to include the noise in the bounds, whereas in [4] it is mainly used to improve the dependence on n when the training error goes to 0.

All in all, there is a wide literature on theoretical bounds on the risk, improving on their interpretability or tightness, which has been a fundamental compass for the mathematical roadmap of this thesis. On the empirical side, several works evaluate such bounds in practice, but with the aim of selecting models or certifying their accuracy, rather than describing and interpreting the generalization behaviour of decision trees. Our work positions itself as a story of theory and evidence aimed at explaining the generalization of decision trees rigorously. To the best of our knowledge, it is the first work that compares the shape of theoretical bounds, as functions of the complexity of the hypotheses set, with the empirical generalization gap of decision trees.

Setting

Given a sample space Ω, let inputs and labels be generated by $X : \Omega \to \mathcal{X} = \{\bar{x}_1, \dots, \bar{x}_N\}$ and $Y : \Omega \to \mathcal{Y} = \{-1, 1\}$. Let p be the joint probability distribution of (X, Y) on $\mathcal{X} \times \mathcal{Y}$, with

$$p_{j,y} = \Pr[X = \bar{x}_j, Y = y], \qquad p_{j,y} \geq 0, \qquad \sum_{j=1}^{N} \sum_{y \in \mathcal{Y}} p_{j,y} = 1.$$

Let $D = (X_i, Y_i)_{i=1}^n$ be a dataset of n i.i.d. copies of (X, Y) drawn from p. We use the term *dataset* to refer to the *random variable* generating the data throughout this work. We denote by $S = (x_i, y_i)_{i=1}^n$ a realization of it. In learning theory, we assume to observe S and hope to learn a function that given a new input $x \in \mathcal{X}$

classifies it correctly according to p. Let $\mathscr{H} \subseteq \{h : \mathcal{X} \to \mathbb{R}\}$ be a nonempty, finite set of predictors where we look for such a function. Given a loss function $\ell : \mathbb{R} \times \mathcal{Y} \to \mathbb{R}^+$, define the population and empirical risks by

$$R(h) = \mathbb{E}_D(\ell(h(X), Y)) = \sum_{\bar{x} \in \mathcal{X}} \sum_{\bar{y} \in \{-1,1\}} \ell(h(\bar{x}), \bar{y}) p(\bar{x}, \bar{y}).$$

and

$$\hat{R}_D(h) = \frac{1}{n} \sum_{i=1}^{n} \ell(h(X_i), Y_i).$$

Because we do not have access to p, we select the predictor that minimizes the empirical risk (ERM), i.e.

$$\hat{h}_D \in \arg\min_{h \in \mathscr{H}} \hat{R}_D(h)$$

which, importantly, is a random variable of the data D. Note that at least one such $\hat{h}_D$ exists, since $\mathscr{H}$ is finite and nonempty.

Approach

We study bounds on the expcted gap between the test and train error of the ERM,

$$\mathbb{E}_D \left[(R - \hat{R}_D)(\hat{h}_D) \right] \tag{1}$$

Note that since $\hat{h}_D$ and $\hat{R}_D$ depend on the same dataset D, $\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \neq R(\hat{h}_D)$, i.e. the empirical risk is a biased estimate of the true risk. A proof of this fact is found in Appendix C.

This approach allows to obtain bounds on the risk of the learnt model $\hat{h}_D$ instead of uniform bounds over the whole space of hypotheses as usual in learning theory, as the algorithm may not explore it [5]. Moreover, the training error usually goes to 0 in our setting where the dataset is smaller than the input space, hence a bound on (1) corresponds to a bound on the test error for a large enough depth. In light of this, we refer to a bound on the gap as a bound on the test error throughout this work. We detach from the usual literature also in that we express concentration results as bounds on the expected value of the target quantity via tail integration, rather than using results in high probability or bounding the probability that this quantity is large. In fact, results in expected value are more practical for the experiments and their interpretation is clearer as compared to high-probability results.

While some results are stated in a general setting, for arbitrary hypothesis classes and loss functions, we develop our main results for binary decision trees under the zero-one loss. Specifically,

- the zero-one loss is $\ell(h(x), y) = \mathbf{1}(h(x) \neq y) = \frac{1-h(x)y}{2}$;
- we consider a number of hypotheses classes $\mathscr{H}$, each denoted by $\mathscr{H}_d$ and corresponding to the set of binary decision trees $h : \mathcal{X} \to \mathcal{Y}$ with depth at most d for $d > 0$, letting d range over increasing values.

Each one of the following sections is organized as follows. First, we present theoretical results in the most generic case given the setting above. Then, we describe the experimental design and state the final asymptotic bound, which we then plot alongside an empirical estimate of (1) to assess how well it captures the qualitative behavior of the test error curve. We conclude commenting on the results.

All proofs are in Appendix A.

2. Union bound over $\mathscr{H}$

In this section, we present a fundamental result on the behavior of the gap between the expected and true risk of the empirical risk minimizer. First, we make the following assumption.

Assumption 2.1 (Boundedness of loss function)**.** *Given a set $A \subseteq \mathbb{R}$ and $\ell : A \to \mathbb{R}^+$, $\exists M \in \mathbb{R}$ s.t.*

$$\ell(a) \leq M \qquad \forall a \in A.$$

A bounded loss function is crucial for obtaining all the upper bounds on the risk discussed in this work. However, the following holds.

Assumption 2.2 (Lipschitzness of loss function and bound at zero)**.** *Given a set $A \subseteq \mathbb{R}$ and $\ell : A \to \mathbb{R}^+$,*

(i) $\exists C > 0$ *s.t.*

$$\forall a, a' \in A, |\ell(a) - \ell(a')| \leq C|a - a'|$$

(ii) $\exists C' > 0$ *s.t.*

$$|\ell(0)| \leq C'$$

Proposition 2.3. *For $A \coloneqq \{a \in \mathbb{R} : |a| \leq B$ for some constant $B \in \mathbb{R}\}$, Assumption 2.2 $\Rightarrow$ Assumption 2.1.*

Note that, in general, we can always take a *truncation map* $T : \mathbb{R} \to [-1, 1]$, defined as

$$T(a) = \begin{cases} -1 & a \leq -1 \\ a & -1 < a < 1, \\ 1 & a \geq 1. \end{cases} \qquad a \in A \tag{2}$$

This allows to set $\mathscr{H}_T = T \circ \mathscr{H}$ and take $A = \{(h(x), y) : h \in \mathscr{H}_T, x \in \mathcal{X}, y \in \mathcal{Y} = \{1, -1\}\}$ which satisfies the assumption of bounded inputs required by Proposition 2.3.

Proposition 2.3 combined with the truncation map argument allows to reduce the requirement of boundedness to that of Lipschitzness of the loss function. The latter does not depend on the hypotheses set and is sometimes easier to obtain, as Example 2.4 shows. Also, this provides a framework for extending the results in this work to an arbitrary $\mathscr{H}$, possibly infinite, as discussed in Section 6.

Example 2.4 (Logistic loss function)**.** *Let $\ell(a) = \log(1+e^{-a}), a \in \mathbb{R}$ be the logistic loss function. Note that in our setting the function takes as input the product $h(x)\cdot y$. We have*

$$\ell'(a) = -\frac{e^{-a}}{1+e^{-a}} = -\frac{1}{1+e^{a}} \quad \Rightarrow \quad |\ell'(a)| < 1 \qquad \forall a \in \mathbb{R}$$

hence

$$C = \sup_{a\in\mathbb{R}} |\ell'| \leq 1.$$

Also,

$$\ell(0) = \log 2 = C'.$$

Remark 2.5. *The zero-one loss satisfies Assumption 2.1 with $M = 1$ by definition.*

We are ready to state our first result. We obtain the upper bound starting from an exponential concentration result (Hoeffding's inequality, Theorem A.1 in the Appendix) applied to the random variable $\hat{R}_D(h)$ for a fixed function $h \in \mathscr{H}$. We then obtain the result at $\hat{h}_D$ through a union bound (Proposition A.4 in the Appendix). This is because $\hat{h}_D$ is not fixed as it depends on the data D, hence we need to bound the probability that the event holds for any function $h \in \mathscr{H}$. This justifies the dependence on $|\mathscr{H}|$. Finally, we tail integrate to obtain a bound on the expected value, which yields the logarithmic rate.

Theorem 2.6. *Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.3. Then*

$$0 \leq \mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_D)] \leq 2M\sqrt{\frac{\log|\mathscr{H}|}{2n}}$$

The result shows that the empirical risk of the empirical risk minimizer underestimates its true risk and it provides a bound on the gap that depends on the square

root of the logarithm of the cardinality of the hypotheses space. We now describe the experimental design for plotting this bound in the case of binary decision trees.

Experimental design

The preliminary experiments mentioned below are described and their plots are shown in Appendix B.

We create the support $\mathcal{X}$ as a discretization over $N = 10,000$ numbers of the interval $[0, 10]$. We sample from it with the uniform measure $p_{\mathcal{X}}(\bar{x}) = \sum_{\bar{y}\in\mathcal{Y}} p(\bar{x}, \bar{y}) = \frac{1}{N}$. The law to generate the labels is

$$y \leftarrow \text{sign}(h^*(x) + \sigma\varepsilon)$$

where $h^* : \mathcal{X} \to \mathcal{Y}$, is the data-generating function, $\varepsilon \sim \mathcal{N}(0, 1)$, is a Gaussian noise, and $\sigma > 0$. First, we generate the labels with a centered parabola, so that roughly half of them are 1 and half -1 in the interval $[0, 10]$. Since an algorithm could learn such a function with few datapoints (see Section B.2), we switch to a sinusoid with varying frequency,

$$h^*(x) = 5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right)$$

where m is the number of times h^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of $1s$ and $-1s$. We set as default value $m = 10$.

We test a Bernoullian noise instead of a Gaussian one, concluding that there is no interesting difference in the performance of the algorithm as long as the portion of misclassified inputs due to the noise is the same. We find that for $m = 10$, $\sigma = 3.5$ achieves a noise of 20%, corresponding to $p = 0.8$. For now, we stick to a gaussian noise.

In the end, our labels are generated as

$$y \leftarrow \text{sign}\left(5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right) + \sigma\varepsilon\right)$$

with $\sigma = 3.5$ and $m = 10$. Also, our sign$(\cdot)$ maps 0 to 1.

We proceed as follows. We generate a test set with size $n_{\text{test}} = 200,000$ sampling from $\mathcal{X}$ with $p_{\mathcal{X}}$ and with replacement, and generating the labels as above. Then,

for each depth $d \in [1, 7\lceil \log_2 n \rceil] \subset \mathbb{N}$ we do the following. For a number $T = 100$ of iterations, we

- generate a train set ($n = 50$);
- fit a decision tree with depth at most d (DecisionTreeClassifier algorithm from the scikit-learn library);
- compute the train error, the test error, and the differene between the two (the gap).

Then, we average the resulting quantities over the T samples, obtaining empirical estimates of the expected values of the train error, test error, and gap over the disorder of the data.

To plot the theoretical bound, instead, we need to express $|\mathscr{H}_d|$ as a function of d. The exact computation gives

$$|\mathscr{H}_d| = 2 \sum_{k=0}^{\min N-1, 2^d-1} \binom{N-1}{k}$$

Disclaimer: all the plots in this thesis have been obtained with $|\mathscr{H}_d| \leq 2^d$ which is incorrect, I need to fix it. I tried the formula above or something like $\min(2^N, N^{2^d} 2^{2^d})$ as suggested by LLMs, but dependence on d is awful and the empirical error curve gets squashed down. However, this is not a problem for the rigor and validity of the results.

As the maximum depth d increases, we plot the following:

1. the train and the test error curves;
2. the gap and the theoretical bound $\sqrt{\log |\mathscr{H}|/n}$;
3. the test error and the sum of train error and the bound (moving $\hat{R}_D(\hat{h}_D)$ to the right).

Considering the zero-one loss and ignoring constants since we are interested in the asymptotic behavior, Theorem 2.6 gives

$$\mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_{D,d})] \leq O\left(\sqrt{\frac{\log |\mathscr{H}_d|}{n}}\right) \qquad \text{for } d > 0. \tag{3}$$

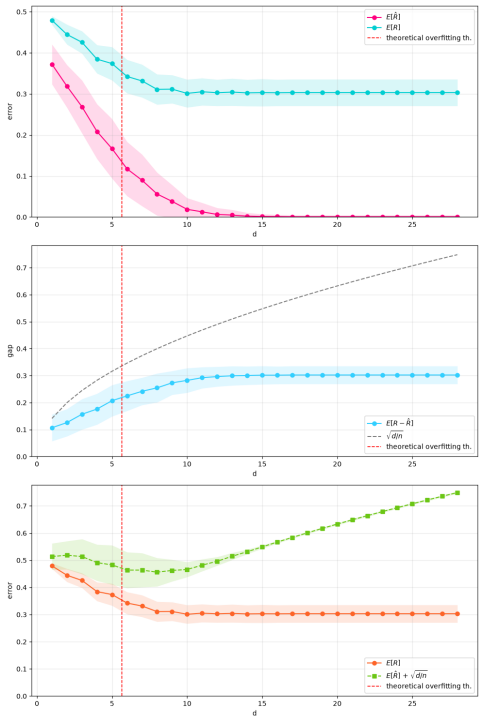


FIGURE 1. Union bound (Theorem 2.6) over $\mathscr{H}_d$. Train and test error, gap against the bound, and test error against train error plus the bound, as functions of the maximum depth d; $n = 50$, $m = 10$, Gaussian noise with $\sigma = 3.5$.

We plot the bound (3) in Figure 1. The first thing catching the eye is that the bound diverges, while the test error saturates. This divergence is due to the dependence of the bound on $|\mathscr{H}_d|$, that is the number of binary trees with maximum depth d, which clearly increases with d. It follows immediately that a complexity measure of the hypotheses set that naively counts the functions it contains does not satisfy our requirements to find a qualitatively informative bound.

The test error of binary decision trees saturates because of a specific characteristic of the algorithm. As soon as the learned tree has only "pure" leaves, i.e. leaves containing only inputs that have the same label in the training set, the algortihm ceases to split them and hence any additional allowed depth does not change the learned model. Clearly, the cardinality of the hypotheses sets $\mathscr{H}_d, d > 0$ does capture this property. Instead, a counting way that accounts for the finiteness of

our problem might reproduce the saturation of the test error curve. Specifically, the configuration with only pure leaves can be achieved for two reasons: either all the possible values that the inputs can take are classified, or all the inputs in the sample are classified. Since, in our case, $n \ll N$, the second event happens much earlier than the first one. As a consequence, a complexity measure that counts the number of distinct classifications that the functions in $\mathscr{H}$ can give, i.e. the vectors $h(x_1), \ldots, h(x_n) : h \in \mathscr{H}$, instead of the functions themselves, could more faithfully represent the true complexity of the space of models that the algorithm can learn in practice.

3. Using the Rademacher complexity

In this section, we introduce the Rademacher complexity, a tool that allows to leverage the viewpoint exposed in the last section and obtain a saturating bound on the test error of decision trees. As before, we state the first results for generic loss functions and hypotheses sets.

Define $\mathscr{L} = \{\ell_h : h \in \mathscr{H}\}$ and $\ell_h(x, y) = \ell(h(x), y)$.

Definition 3.1 (Rademacher complexity)**.** *Let $\xi_1, \dots, \xi_n$ be independent random variables taking value 1 or -1 with equal probability. The Rademacher complexity of the set $\mathscr{L}$ is*

$$\mathcal{R}_n(\mathscr{L}) \coloneqq \mathbb{E}_{\xi, D}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i)\right]$$

The Rademacher complexity measures the richness of a class of functions. The argument of the supremum $\frac{1}{n}\sum_i \xi_i \ell_h(X_i, Y_i) = \langle \xi, \ell_h(X, Y)\rangle$ is the correlation between ξ and $\ell_h(X, Y)$. Since the loss function is fixed, this correlation measures to what extent the predictors can align to random labels. Note that the supremum is taken before the expectation over the disorder of ξ and D, and that in our case it is a maximum, since $\mathscr{L}$ is finite. Hence, for each dataset and for each realization of the Rademacher vectors, we can choose a different function $\ell_h \in \mathscr{L}$ that attains the maximum. A high complexity means that for most values of ξ we can find a model h that is similar to that pattern. It is the highest when we can find a function that correlates with any possible pattern.

The following results hold.

Define $N_D(\mathscr{H}) = |\{(\ell_h(X_1, Y_1), \dots, \ell_h(X_n, Y_n)) : h \in \mathscr{H}\}|$.

Theorem 3.2 (Rademacher complexity bound)**.** *[[1], Proposition 4.2]*

$$\mathbb{E}_D\left[\sup_{h \in \mathscr{H}} (R - \hat{R}_D)(h)\right] \leq 2\mathcal{R}_n(\mathscr{L})$$

Lemma 3.3 (Massart)**.** *[[14], Theorem 3.3] Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.3. Then*

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log|\mathscr{H}|}{n}}$$

Note that Theorem 3.2 can be easily extended to a bound on the gap at $\hat{h}_D$ because

$$\mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_D)] \leq \mathbb{E}_D\left[\sup_{h \in \mathscr{H}} (R - \hat{R}_D)(h)\right]$$

since $\hat{h}_D \in \mathscr{H}$. Moreover, the supremum in Definition 3.1 can be equivalently computed over the set of loss vectors, which yields the following version of the lemma.

Lemma 3.4 (Massart, loss vectors)**.** *Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.3. Then*

$$\mathcal{R}_n(\mathscr{L}) \leq \mathbb{E}_D\left[M\sqrt{\frac{2\log N_D(\mathscr{H})}{n}}\right]$$

So, using the Rademacher complexity as an intermediate step allows us to replace the number of predictors $|\mathscr{H}|$ from Theorem 2.6 with the number of loss vectors $N_D(\mathscr{H})$ (up to constants). In fact, Theorem 3.2 bounds a quantity that depends on the true expected value of $\ell_h(X, Y)$ with a quantity that can be computed on loss vectors. The key ingredient for proving it is a symmetrization technique, which consists in replacing the deviation of the random variable $\ell_h(X, Y)$ from its expected value with the difference between two identical random variables taking independent datasets D and D' as inputs. Note that since the loss function is fixed, counting loss vectors is equivalent to counting classification vectors. We work with loss vectors here because they arise more naturally from Rademacher complexity, and turn to classification vectors in Section 5.

The following result encloses the key point of the saturation mechanism discussed at the end of Section 2.

Lemma 3.5 (Counting inequalities)**.** *Let ℓ be the zero-one loss. Then*

$$N_D(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$$

surely.

Substituting this bound on N_D in Lemma 3.4 and combining it with Theorem 3.2, we obtain the following bound.

Theorem 3.6 (Rademacher and Massart bound)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_D)] \leq \min\left\{2\sqrt{\frac{2\log|\mathscr{H}|}{n}}, 2\sqrt{2\log 2}\right\}$$

The experimental design is the same as above. Again, we ignore the constants and use

$$\min\left\{2\sqrt{\frac{2\log|\mathscr{H}|}{n}}, 2\sqrt{2\log 2}\right\} = \min\left\{O\left(\sqrt{\frac{\log|\mathscr{H}|}{n}}\right), O(1)\right\}. \tag{4}$$

We set the constant term to 0.5, because the Rademacher complexity of $\mathscr{L}$ is naturally bounded by 0.5 (see Appendix C for a proof of this fact).

Related to the comment on the formula for $|\mathscr{H}|$, also the constants we choose need to be revisited. Again, saturation holds nonetheless. It's a matter of how to display the results.

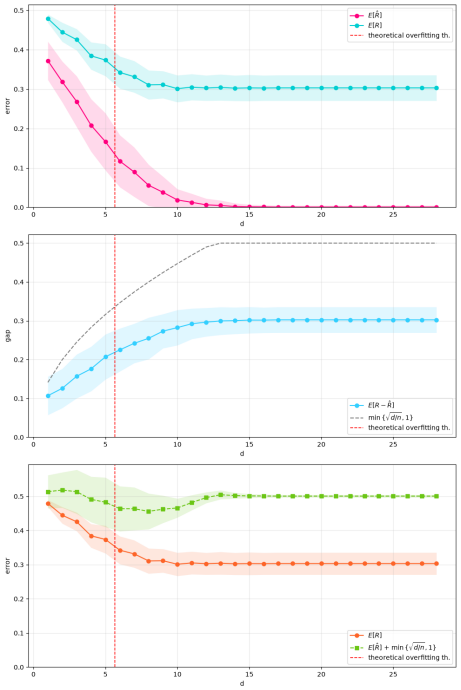


FIGURE 2. Rademacher and Massart bound (Theorem 3.6), same setting as Figure 1.

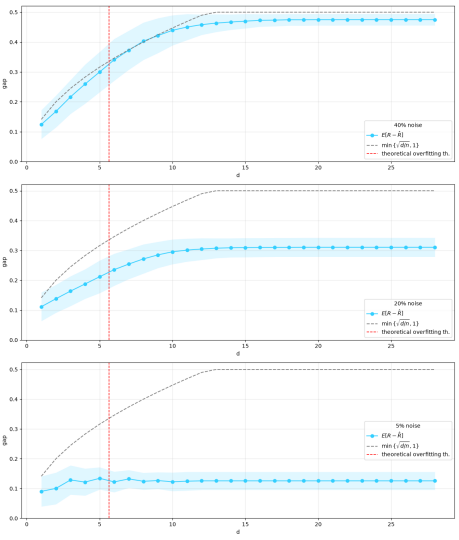


FIGURE 3. Rademacher and Massart bound against the gap for different noise levels.

We plot bound 4 against the empirical error curve in Figure 2. In the leften part of the graph, the bound is the same as in Figure 1. From a depth of 13 onwards, the minimum is instead attained by the constant term, hence the bound saturates. This reflects the fact that beyond a certain depth, the number of distinct classification vectors no longer increases with d, through the mechanism explained above.

At the same time, this mechanism is myopic to the other aspects of the structure of the learning problem. Indeed, Figure 3 shows how the gap curve is sensistive to different levels of noise in the dataset, as measured by the σ coefficient (standard deviation). Specifically, it remains flatter with less noise, indicating both that the model performs better and that the difference between train and test error is lower (see the explorative experiments in Appendix B showing how the empirical curves change with noise). We use $\sigma \in \{12.5, 3.5, 1\}$, which flip on average 40%, 20% and 5% of the labels, the same noise levels as $p \in \{0.6, 0.8, 0.95\}$ in Section 4. The bound curve, instead, remains unchanged, failing to capture that the noise present in the data has an effect on the generalization performance of the algorithm.

4. Dependence on the best risk

In this section, we implement a different approach and new theoretical tools to look for a noise-sensitive bound.

The first ingredient is a concentration result for $\hat{R}_D$ that depends on the true risk R.

Define $N_n(\mathscr{H}) := \sup_{S \in (\mathcal{X} \times \mathcal{Y})^n} N_S(\mathscr{H})$. Let ℓ be the zero-one loss for the rest of this section.

Lemma 4.1 (Relative deviation)**.** *For $\varepsilon > 0$ with $n\varepsilon^2 \geq 8$,*

$$\Pr\left[\exists h \in \mathscr{H} : (R - \hat{R}_D)(h) > \varepsilon\sqrt{R}(h)\right] \leq 2N_{2n}(\mathscr{H})e^{-n\varepsilon^2/16}.$$

The dependence on the supremum of the number of loss vectors N_n emerges naturally, since the proof of this result uses a symmetrization technique. Using the supremum instead of the expectation over D as in Lemma 3.4 is only a trick to avoid handling the dependence on the disorder of the dataset.

Remark 4.2. *Lemma 4.1 uses Chebyshev inequality that bounds the deviation of a random variable in terms of its variance. For a bounded random variable, the variance is controlled by the first moment, following the intuition that if being bounded it cannot fluctuate much around its expected value. Specifically, if Z is Bernoulli, then $\mathrm{Var}(Z) = \mathbb{E}[Z]\,(1 - \mathbb{E}[Z]) \leq \mathbb{E}[Z]$. Through the lenses of learning theory, for $Z = \ell_h(X, Y)$, si ha*

$$\mathrm{Var}(\ell_h) \leq R(\ell_h(X, Y))$$

and hence a bound in terms of the best risk. Then, applying a union bound, the resuly above follows (as always, the complete proof is in Appendix A).

In high probability, and isolating the empirical risk $\hat{R}_D$ on the right hand side, we obtain the following result.

Theorem 4.3 (Bound with empirical risk)**.** *With probability at least $1 - \delta$,*

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2N_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n}$$

Let the minimum expected risk achievable in the hypotheses set be

$$R^*_{\mathscr{H}} = \min_{h \in \mathscr{H}} R(h).$$

and let

$$h_b \in \arg\min_{h \in \mathscr{H}} R(h)$$

be the best hypothesis, i.e. the model attaining $R^*_{\mathscr{H}}$. Again, at least one such model exists since $\mathscr{H}$ is finite and nonempty.

By definition of empirical risk minimizer, we know that its empirical risk is controlled by the empirical risk attained by the best hypotheses h_b. Therefore, controlling $\hat{R}(h_b)$ yields a bound on the empirical risk of the ERM that can be used in Theorem 4.3.

To do so, we use Bernstein inequality (see Theorem A.3 in the Appendix) on the variable $\hat{R}_D(h_b)$. We obtain a bound on $\hat{R}_D(\hat{h}_b)$ that depends on the best risk $\mathbb{E}_D[\hat{R}_D(h_b)] = R^*_{\mathscr{H}}$. Substituting into the previous result, we get a bound on the gap at the ERM that depends on the best risk.

Corollary 4.4 (Bound with best risk)**.** *With probability at least* $1 - \delta$*,*

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{R^*_{\mathscr{H}} \log(4N_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}$$

Note that Bernstein inequality used to obtain this result is a refinement of Chebyshev's [**?**]. Indeed, this result uses the same trick explained in Remark 4.2 obtaining $\mathrm{Var}(h_b) \leq R(h_b) = R^*_{\mathscr{H}}$.

Let the Bayes risk be

$$R^* := \inf_{h : \mathcal{X} \to \mathcal{Y}} R(h)$$

where the infimum runs over a superset of $\mathscr{H}$. The best risk can be decomposed as

$$R^*_{\mathscr{H}} = \underbrace{R^*}_{\text{Bayes risk}} + \underbrace{R^*_{\mathscr{H}} - R^*}_{\text{Approximation error}} =: \sigma^2 + A_{\mathscr{H}}$$

where

- the Bayes risk σ^2 measures the classification error due to the irreducible noise ε, as it is the minimal error that any function $h : \mathcal{X} \to \mathcal{Y}$ necessarily makes;

· the approximation error $A_{\mathscr{H}}$ is the excess error we make given that we consider a restricted set of predictors, $\mathscr{H}$.

Note that if there is no noise, then $R^* = 0$ and the Bayes risk is attained by the data-generating function h^*. Moreover, if the model attaining R^* lives in the hypotheses class, then $A_{\mathscr{H}} = 0$.

Corollary 4.5. *With probability* $1 - \delta$,

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\log(4N_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}$$

Tail integrating and using the counting Lemma 3.5, we obtain the following result.

Theorem 4.6 (Noise and approximation error bound)**.** *Let* ℓ *be the zero-one loss. Then*

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \leq \min\left\{4\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\,(\log(4|\mathscr{H}|) + 1)}{n}} + 20\,\frac{\log(4|\mathscr{H}|) + 1}{n},\right.$$
$$\left. 4\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\,((2n+2)\log 2 + 1)}{n}} + 20\,\frac{(2n+2)\log 2 + 1}{n}\right\}$$

Remark 4.7. *All in all, we have used that the empirical risk of the ERM is never worse than the empirical risk of the best hypothesis. This, in turn, concentrates around its true risk, the best risk. The best risk achievable by our hypotheses is determined by the unavoidable risk any function* $h : \mathcal{X} \to \mathcal{Y}$ *would make due to noise in the data, plus an excess risk due to our choice of hypotheses. This naturally yields a bound depending on the noise.*

Experimental design

As compared to Sections 2 and 3 that used a Gaussian noise, we switch to a label-flip noise with flip probability $1 - p$. Therefore, we sample from $\mathcal{X}$ as before and we generate the labels as

$$y \leftarrow \operatorname{sign}(h^*(x)) \cdot b$$

where h^* is as above and b is a random variable taking value 1 with probability p and -1 with probability $1 - p$. We use $p = 0.8$ and $p \in \{0.6, 0.8, 0.95\}$ to inspect multiple levels of noise.

Since $p \geq 0.5$, the Bayes risk is

$$R^* = 1 - p$$

$R^*_{\mathscr{H}}$ is computed exactly using dynamic programming. We produce labels using our sine function and the Bernoulli error and we cluster the intervals where labels have the same sign, obtaining L contiguous blocks. We create a dp $k \times 2$ matrix that stores the minimum risk for each number of maximum splits k, up to $\min(L-1, 2^d)$ where d is the maximum depth we are considering, when the last assigned label is 1 or -1 (the columns). Then, for each contiguous block we dynamically update the dp matrix summing the minimum risk among all trees with at most k splits. In the end, we select the minimum risk among the tree predicting 1 or -1 in the very last block.

The rest of the experimental design is as above.

As before, we ignore the constants and plot

$$\min \left\{ O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2) \log |\mathscr{H}|}{n}} + \frac{\log |\mathscr{H}|}{n} \right), O\left(\sqrt{A_{\mathscr{H}} + \sigma^2} + 1 \right) \right\}. \quad (5)$$

We plot bound (5) in Figure 4. As before, the bound saturates due to the constant term. Figure 5 shows the two terms in the minimum. Figure 6 shows the bound and the test error curve for different noise levels. Finally, we see that the bound changes with the noise. Both the term depending on d and the constant one, which determines the saturating level, depend on the minimum risk $R^*_{\mathscr{H}}$. As explained above, this changes with the noise, hence the bound capture the sensitivity of the generalization performance to different noise levels. This is shown in Figure 7.

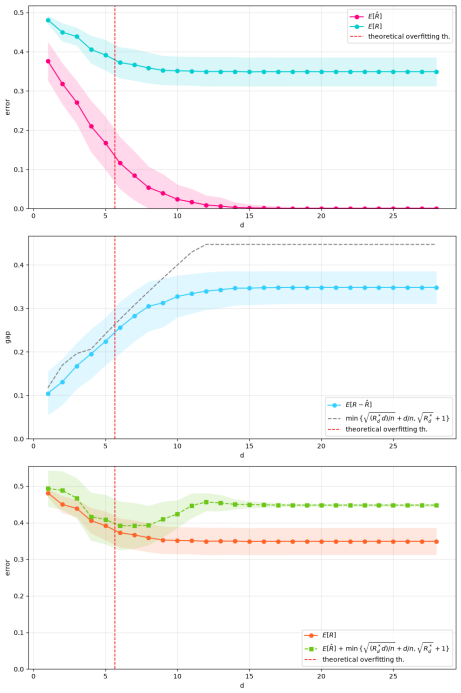


FIGURE 4. Noise and approximation error bound (Theorem 4.6), same setting as Figure 1 with label-flip noise at 20% ($p = 0.8$).

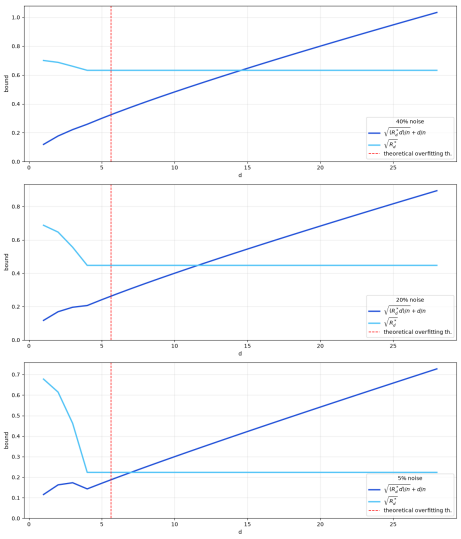


FIGURE 5. The two terms in the minimum of the noise and approximation error bound, for different noise levels.

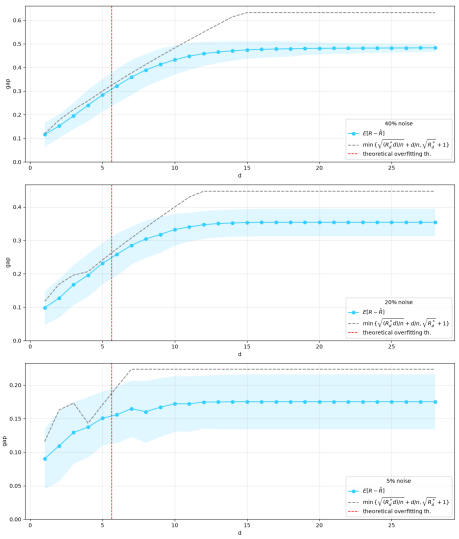


FIGURE 6. Noise and approximation error bound against the gap, for different noise levels.

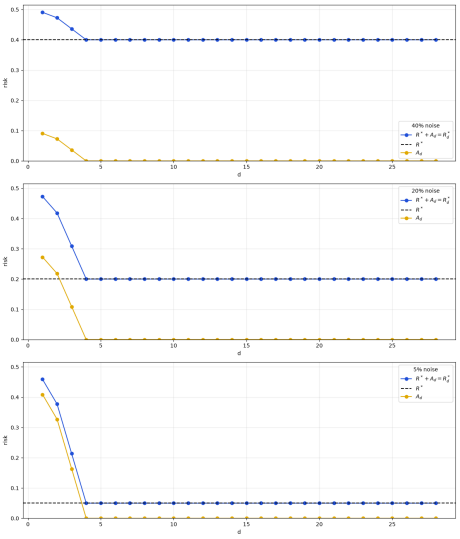


FIGURE 7. Best risk $R^*_{\mathscr{H}_d}$ split into Bayes risk $R^* = 1 - p$ and approximation error $A_{\mathscr{H}_d}$, for different noise levels.

5. Growth function

In this section, we introduce a new measure of the complexity of the hypotheses class, that considers classification vectors. This tool allows to make the intuition on the saturation of the test error due to characteristics of the decision tree algorithm that we discussed in Section 3 more explicit.

Definition 5.1 (Growth function)**.** *The growth function of $\mathscr{H}$ is*

$$\Pi_n(\mathscr{H}) := \sup_{x_1,\dots,x_n \in \mathcal{X}} \left|\{(h(x_1),\dots,h(x_n)) : h \in \mathscr{H}\}\right|.$$

The following chain of inequalities extends Lemma 3.5. Given the set of images of a function h, applying a loss function to it can at most decrease the number of vectors in the set. At the same time, the images of the functions in $\mathscr{H}$ are at most as many as the functions in $\mathscr{H}$, and always less than or equal to 2^n, the number of all possible n-long sequences of 1 and -1.

The following extends Lemma 3.5.

Lemma 5.2 (Counting inequalities with growth function)**.** *Let ℓ be the zero-one loss. Then*

$$1 \leq N_D(\mathscr{H}) \leq \Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$$

surely. Moreover,

$$\Pi_n(\mathscr{H}) = N_n(\mathscr{H}) = \sup_{S \in (\mathcal{X}\times\mathcal{Y})^n} N_S(\mathscr{H}).$$

The relation between N_D and Π_n makes it immediate to obtain growth-function versions of the bounds in Sections 3 and 4 (substitute $N(D) \leq \Pi_n$ into Lemma 3.4 and $N_{2n} = \Pi_{2n}$ into Theorem 4.3). Substituting also $\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$ yields bounds (4) and (5). The bounds are the same, but the intuition that they saturate because the number of distinct classification vectors is at most 2^n is made more explicit. In other words, when we evaluate the performance of an algorithm, the models in $\mathscr{H}$ that the algorithm can learn are equivalent if they give the same classification. This result takes this into account when measuring the richness of the hypotheses class $\mathscr{H}$.

The following Theorem constitutes our final result. It provides both an intuition on the drivers of the test error of decision trees through the dependence on Π_{2n}, A, σ^2, and captures its shape empirically as shown in Figure 4.

Theorem 5.3 (Bound with growth function)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}_D\left[(R-\hat{R}_D)(\hat{h}_D)\right] \leq 4\sqrt{\frac{(A_{\mathscr{H}}+\sigma^2)\left(\log(4\Pi_{2n}(\mathscr{H}))+1\right)}{n}}+20\,\frac{\log(4\Pi_{2n}(\mathscr{H}))+1}{n}$$

with

$$\Pi_{2n}(\mathscr{H}) \leq \min\left\{|\mathscr{H}|, 2^{2n}\right\}.$$

6. Conclusions

finallyyy

References

[1] Francis Bach. *Learning theory from first principles*. MIT Press, 2024.

[2] Peter L. Bartlett and Shahar Mendelson. Rademacher and Gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3(Nov):463–482, 2002.

[3] Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine-learning practice and the classical bias–variance trade-off. *Proceedings of the National Academy of Sciences*, 116(32):15849–15854, 2019.

[4] Stéphane Boucheron, Olivier Bousquet, and Gábor Lugosi. Theory of classification: A survey of some recent advances. *ESAIM: Probability and Statistics*, 9:323–375, 2005.

[5] Olivier Bousquet and André Elisseeff. Stability and generalization. *Journal of Machine Learning Research*, 2(Mar):499–526, 2002.

[6] Alicia Curth, Alan Jeffares, and Mihaela van der Schaar. A u-turn on double descent: Rethinking parameter counting in statistical learning. *Advances in Neural Information Processing Systems*, 36:55932–55962, 2023.

[7] Mostefa Golea, Peter L. Bartlett, Wee Sun Lee, and Llew Mason. Generalization in decision trees and DNF: Does size matter? In *Advances in Neural Information Processing Systems*, volume 10, pages 259–265, 1998.

[8] Yiding Jiang, Behnam Neyshabur, Hossein Mobahi, Dilip Krishnan, and Samy Bengio. Fantastic generalization measures and where to find them. In *International Conference on Learning Representations*, 2020.

[9] Matti Kääriäinen and John Langford. A comparison of tight generalization error bounds. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 409–416, 2005.

[10] Matti Kääriäinen, Tuomo Malinen, and Tapio Elomaa. Selective Rademacher penalization and reduced error pruning of decision trees. *Journal of Machine Learning Research*, 5(Sep):1107–1126, 2004.

[11] Jason M. Klusowski and Peter M. Tian. Large scale prediction with decision trees. *Journal of the American Statistical Association*, 119(545):525–537, 2024.

[12] Jean-Samuel Leboeuf, Frédéric LeBlanc, and Mario Marchand. Generalization properties of decision trees on real-valued and categorical features. *arXiv preprint arXiv:2210.10781*, 2022.

[13] Yishay Mansour and David A. McAllester. Generalization bounds for decision trees. In *Proceedings of the Thirteenth Annual Conference on Computational Learning Theory (COLT)*, pages 69–74, 2000.

[14] Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. *Foundations of machine learning*. MIT Press, 2018.

[15] Vaishnavh Nagarajan and J. Zico Kolter. Uniform convergence may be unable to explain generalization in deep learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[16] Chaithanya Pichuka, Raju S. Bapi, Chakravarthy Bhagvati, Arun K. Pujari, and Bulusu Lakshmana Deekshatulu. A tighter error bound for decision tree learning using PAC learnability. In *Proceedings of the International Joint Conference on Artificial Intelligence*, pages 1011–1016, 2007.

[17] Vladimir N. Vapnik. *The nature of statistical learning theory*. Springer Science & Business Media, 2013.

[18] Olcay Taner Yıldız. VC-dimension of univariate decision trees. *IEEE Transactions on Neural Networks and Learning Systems*, 26(2):378–387, 2015.

[19] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning requires rethinking generalization. In *International Conference on Learning Representations*, 2017.

[20] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 64(3):107–115, 2021.

Appendix A. Proofs

We first recall some basic results that are used throughout, and then give the proofs of the results in the main text.

Theorem A.1 (Hoeffding's inequality)**.** *Let $Z_1, \dots, Z_n$ be independent real-valued random variables with $Z_i \in [a_i, b_i]$ almost surely for some $a_i, b_i \in \mathbb{R}$, and let $\mu = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[Z_i]$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^n Z_i - \mu \geq \varepsilon\right] \leq \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Theorem A.2 (Chebyshev's inequality)**.** *Let Z be a real-valued random variable with finite variance. Then, for every $\varepsilon > 0$,*

$$\Pr\left[|Z - \mathbb{E}[Z]| \geq \varepsilon\right] \leq \frac{\mathrm{Var}(Z)}{\varepsilon^2}.$$

Theorem A.3 (Bernstein's inequality)**.** *Let $Z_1, \dots, Z_n$ be independent real-valued random variables with $\mathbb{E}[Z_i] = 0$ and $Z_i \leq M$ almost surely for some $M > 0$, and let $v = \frac{1}{n}\sum_{i=1}^n \mathrm{Var}(Z_i)$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^n Z_i \geq \varepsilon\right] \leq \exp\left(-\frac{n\varepsilon^2}{2(v + M\varepsilon/3)}\right).$$

Proposition A.4 (Union bound)**.** *For any finite or countable collection of events $A_1, A_2, \dots$,*

$$\Pr\left[\bigcup_i A_i\right] \leq \sum_i \Pr[A_i].$$

Theorem A.5 (Hoeffding's lemma)**.** *Let Z be a real-valued random variable with $\mathbb{E}[Z] = 0$ and $|Z| \leq a$ almost surely for some $a \in \mathbb{R}$. Then, for every $t \in \mathbb{R}$,*

$$\mathbb{E}\left[e^{tZ}\right] \leq e^{t^2 a^2/2}.$$

Proposition A.6 (Tail integration)**.** *For any non-negative random variable Z,*

$$\mathbb{E}[Z] = \int_0^\infty \Pr[Z \geq \varepsilon]\ \mathrm{d}\varepsilon.$$

Proposition A.7 (Gaussian tail bound)**.** *For every $c > 0$ and $x_0 > 0$,*

$$\int_{x_0}^\infty e^{-cx^2}\ \mathrm{d}x \leq \frac{1}{2cx_0} e^{-cx_0^2}.$$

Proof of Proposition 2.3. Let $\ell : A \to \mathbb{R}^+$ satisfy Assumption 2.2 with constants C and C'. Then

$$\begin{aligned}\ell(a) &= |\ell(a)| \\ &= |\ell(a) - \ell(0) + \ell(0)| \\ &\leq |\ell(a) - \ell(0)| + |\ell(0)| \\ &\leq C|a| + C' \\ &\leq CB + C' =: M\end{aligned}$$

□

Proof of Theorem 2.6. We prove the lower bound first. Since $\hat{h}_D \in \arg\min_{h \in \mathscr{H}} \hat{R}_D(h)$, we have

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h) \qquad \forall h \in \mathscr{H},$$

and taking expectations

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq \mathbb{E}_D[\hat{R}_D(h)] \qquad \forall h \in \mathscr{H}.$$

For a fixed $h \in \mathscr{H}$, which does not depend on D, the empirical risk is an unbiased estimate of the true risk, since by linearity and because the (X_i, Y_i) are distributed as (X, Y),

$$\mathbb{E}_D[\hat{R}_D(h)] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\ell(h(X_i), Y_i)\right] = R(h).$$

Hence

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq R(h) \qquad \forall h \in \mathscr{H}.$$

Now take S a realization of D and consider in particular $h = \hat{h}_S \in \mathscr{H}$,

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq R(\hat{h}_S).$$

Since this holds for every realization S of D, we have $\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq R(\hat{h}_D)$ almost surely and, taking expectations,

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq \mathbb{E}_D[R(\hat{h}_D)].$$

Next, we prove the upper bound. Fix $h \in \mathscr{H}$ and let $Z_i \coloneqq -\ell(h(X_i), Y_i)$, $i = 1, \dots, n$, so that

$$\frac{1}{n}\sum_{i=1}^{n} Z_i - \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Z_i] = (R - \hat{R}_D)(h).$$

We want to apply Hoeffding's inequality (Theorem A.1). The Z_i are independent, since the (X_i, Y_i) are, and $Z_i \in [-M, 0]$ by non-negativity of the loss function and Proposition 2.3. Hence, with $a_i = -M$ and $b_i = 0$, we obtain

$$\Pr\left[(R - \hat{R}_D)(h) \geq \varepsilon\right] \leq e^{-2n\varepsilon^2/M^2}. \tag{6}$$

Now, to obtain a bound at $\hat{h}_D$, which depends on D, notice that if $(R - \hat{R}_D)(\hat{h}_D) \geq \varepsilon$ then $(R - \hat{R}_D)(h) \geq \varepsilon$ for some $h \in \mathscr{H}$. Therefore, by the union bound (Proposition A.4) and (6),

$$\begin{aligned}
\Pr\left[(R - \hat{R}_D)(\hat{h}_D) \geq \varepsilon\right] &\leq \Pr\left[\bigcup_{h \in \mathscr{H}} \left\{(R - \hat{R}_D)(h) \geq \varepsilon\right\}\right] \\
&\leq \sum_{h \in \mathscr{H}} \Pr\left[(R - \hat{R}_D)(h) \geq \varepsilon\right] \\
&\leq |\mathscr{H}| e^{-2n\varepsilon^2/M^2}.
\end{aligned} \tag{7}$$

To obtain a bound in expected value from (7), we tail integrate. Let $X \coloneqq (R - \hat{R}_D)(\hat{h}_D)$ and $X^+ \coloneqq \max\{X, 0\}$, which is non-negative, so that by Proposition A.6

$$\mathbb{E}_D[X] \leq \mathbb{E}_D[X^+] = \int_0^\infty \Pr[X^+ \geq \varepsilon] \ \mathrm{d}\varepsilon = \int_0^\infty \Pr[X \geq \varepsilon] \ \mathrm{d}\varepsilon.$$

Let $\alpha \coloneqq |\mathscr{H}|$ and $c \coloneqq 2n/M^2$, so that (7) reads $\Pr[X \geq \varepsilon] \leq \alpha e^{-c\varepsilon^2}$. This bound is good for large values of ε, but for small ε it exceeds 1, hence we use instead

$$\Pr[X \geq \varepsilon] \leq \min\{1, \alpha e^{-c\varepsilon^2}\}.$$

We split the integral at the threshold ε_0 where $\alpha e^{-c\varepsilon_0^2} = 1$, i.e. $\varepsilon_0 = \sqrt{\log \alpha / c}$. Using the Gaussian tail bound (Proposition A.7) for the second integral, we obtain

$$\mathbb{E}_D[X] \leq \int_0^{\varepsilon_0} 1 \ \mathrm{d}\varepsilon + \int_{\varepsilon_0}^\infty \alpha e^{-c\varepsilon^2} \ \mathrm{d}\varepsilon \leq \varepsilon_0 + \frac{\alpha}{2c\varepsilon_0} e^{-c\varepsilon_0^2} = \sqrt{\frac{\log \alpha}{c}} + \frac{1}{2\sqrt{c \log \alpha}},$$

where in the last equality we used $\alpha e^{-c\varepsilon_0^2} = 1$. Finally, assuming $\alpha \geq e^{1/2}$, which holds as soon as $|\mathscr{H}| \geq 2$, we have $\frac{1}{2\sqrt{c \log \alpha}} \leq \sqrt{\frac{\log \alpha}{c}}$, and hence

$$\mathbb{E}_D[X] \leq 2\sqrt{\frac{\log \alpha}{c}} = 2M\sqrt{\frac{\log |\mathscr{H}|}{2n}}.$$

□

Proof of Theorem 3.2. Let $Z \coloneqq (X, Y)$ and $Z_i \coloneqq (X_i, Y_i)$, so that

$$R(h) = \mathbb{E}[\ell_h(Z)] \qquad \text{and} \qquad \hat{R}_D(h) = \frac{1}{n}\sum_{i=1}^{n} \ell_h(Z_i).$$

The key ingredient of the proof is to introduce an independent copy $D' = (Z'_1, \dots, Z'_n)$ of the dataset D, and to study the random variables $\ell_h(Z'_i) - \ell_h(Z_i)$, which have a symmetric distribution (*symmetrization trick*). Since the Z'_i are distributed as Z and are independent of D,

$$R(h) = \mathbb{E}_{D'}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i)\right] = \mathbb{E}_{D'}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i) \,\middle|\, D\right].$$

Then,

$$\mathbb{E}_D\left[\sup_{h\in\mathscr{H}} (R - \hat{R}_D)(h)\right] = \mathbb{E}_D\left[\sup_{h\in\mathscr{H}} \mathbb{E}_{D'}\left[\frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right) \,\middle|\, D\right]\right].$$

Since the supremum of expectations is at most the expectation of the supremum, and by the law of total expectation, we obtain

$$\mathbb{E}_D\left[\sup_{h\in\mathscr{H}} (R - \hat{R}_D)(h)\right] \le \mathbb{E}_{D,D'}\left[\sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right)\right]. \tag{8}$$

Now we apply the symmetrization trick. Let $\mathscr{H} = \{h_1, \dots, h_m\}$, with $m = |\mathscr{H}|$. For each $i = 1, \dots, n$, since Z_i and Z'_i are independent and identically distributed, swapping them does not change the joint distribution, i.e.

$$\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \overset{d}{=} \left(\ell_{h_j}(Z_i) - \ell_{h_j}(Z'_i)\right)_{j=1}^{m}.$$

This means that, for any fixed $e = (e_1, \dots, e_n) \in \{-1, 1\}^n$,

$$\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \overset{d}{=} \left(e_i\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

Since the pairs (Z_i, Z'_i) are independent across i, we have

$$\left(\frac{1}{n}\sum_{i=1}^{n}\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m} \overset{d}{=} \left(\frac{1}{n}\sum_{i=1}^{n} e_i\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m},$$

and, since the supremum over $\mathscr{H}$ is a function of this vector,

$$\sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right) \overset{d}{=} \sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} e_i\left(\ell_h(Z'_i) - \ell_h(Z_i)\right).$$

Hence, the expected value of the left-hand side is equal to that of the right-hand side for every $e \in \{-1,1\}^n$. Averaging over e with the distribution of $\xi = (\xi_1, \dots, \xi_n)$, which is independent of D and D', we obtain

$$\mathbb{E}_{D,D'}\left[\sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right] = \mathbb{E}_{\xi,D,D'}\left[\sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} \xi_i\left(\ell_h(Z_i') - \ell_h(Z_i)\right)\right].$$

Plugging this into (8) and bounding the supremum of a sum with the sum of the suprema,

$$\begin{aligned}\mathbb{E}_D\left[\sup_{h\in\mathscr{H}} (R - \hat{R}_D)(h)\right] &\le \mathbb{E}_{\xi,D'}\left[\sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} \xi_i \ell_h(Z_i')\right] + \mathbb{E}_{\xi,D}\left[\sup_{h\in\mathscr{H}} \frac{1}{n}\sum_{i=1}^{n} (-\xi_i)\ell_h(Z_i)\right] \\ &= 2\mathcal{R}_n(\mathscr{L}),\end{aligned}$$

because D' has the same distribution as D, and $-\xi$ has the same distribution as ξ and is independent of D. □

Proof of Lemma 3.3. Let $Z_i \coloneqq (X_i, Y_i)$. For any $t > 0$,

$$\begin{aligned}\exp\left(t\,\mathbb{E}_{\xi,D}\left[\sup_{h\in\mathscr{H}} \sum_{i=1}^{n} \xi_i \ell_h(Z_i)\right]\right) &\le \mathbb{E}_{\xi,D}\left[\exp\left(t \sup_{h\in\mathscr{H}} \sum_{i=1}^{n} \xi_i \ell_h(Z_i)\right)\right] \\ &= \mathbb{E}_{\xi,D}\left[\sup_{h\in\mathscr{H}} \exp\left(t\sum_{i=1}^{n} \xi_i \ell_h(Z_i)\right)\right] \\ &\le \mathbb{E}_{\xi,D}\left[\sum_{h\in\mathscr{H}} \exp\left(t\sum_{i=1}^{n} \xi_i \ell_h(Z_i)\right)\right] \\ &= \mathbb{E}_D\left[\sum_{h\in\mathscr{H}} \prod_{i=1}^{n} \mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i \ell_h(Z_i)\right)\right]\right], \qquad (9)\end{aligned}$$

where we use Jensen's inequality for the convex function exp in the first inequality, that exp is increasing in the first equality, we bound the supremum with the sum (all terms are positive) in the second inequality, and we use the independence of the ξ_i's, and their independence of D, in the last equality.

Now we apply Hoeffding's lemma (Theorem A.5) to each factor, conditionally on D. We can do so because, for fixed Z_i, the random variable $\xi_i \ell_h(Z_i)$ is centered, since $\mathbb{E}[\xi_i] = 0$, and it satisfies $|\xi_i \ell_h(Z_i)| = \ell_h(Z_i) \le M$ by non-negativity of the loss and Proposition 2.3. We obtain

$$\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i \ell_h(Z_i)\right)\right] \le e^{t^2 M^2/2}.$$

Plugging this into (9),

$$\exp\left(t\,\mathbb{E}_{\xi,D}\left[\sup_{h\in\mathscr{H}}\sum_{i=1}^{n}\xi_i\ell_h(Z_i)\right]\right)\leq\sum_{h\in\mathscr{H}}\prod_{i=1}^{n}e^{t^2M^2/2}=|\mathscr{H}|\,e^{nt^2M^2/2}.$$

Taking the logarithm of both sides,

$$t\,\mathbb{E}_{\xi,D}\left[\sup_{h\in\mathscr{H}}\sum_{i=1}^{n}\xi_i\ell_h(Z_i)\right]\leq\log|\mathscr{H}|+\frac{nt^2M^2}{2},$$

and dividing by $tn>0$,

$$\mathcal{R}_n(\mathscr{L})\leq\frac{\log|\mathscr{H}|}{tn}+\frac{tM^2}{2}. \tag{10}$$

Since (10) holds for every $t>0$, we look for the tightest bound. Setting the derivative in t to zero,

$$-\frac{\log|\mathscr{H}|}{t^2n}+\frac{M^2}{2}=0\quad\Longrightarrow\quad t^\star=\sqrt{\frac{2\log|\mathscr{H}|}{nM^2}},$$

and the second derivative $\frac{2\log|\mathscr{H}|}{t^3n}\geq 0$, hence $t^\star$ is a minimizer (if $|\mathscr{H}|=1$, the bound follows by letting $t\to\infty$). Plugging $t^\star$ into (10) yields

$$\mathcal{R}_n(\mathscr{L})\leq M\sqrt{\frac{2\log|\mathscr{H}|}{n}}.$$

□

Proof of Lemma 3.4. Let $Z_i\coloneqq(X_i,Y_i)$. Since ξ is independent of D, by the tower law we can write

$$\mathcal{R}_n(\mathscr{L})=\mathbb{E}_D\left[\mathbb{E}_\xi\left[\sup_{h\in\mathscr{H}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i)\right]\right],$$

where the inner expectation is over ξ only, with D fixed. We bound the inner expectation for every realization S of D, and then take the expectation over D. This is the reason why the expectation over D remains outside of the bound: the number of loss vectors $N_D(\mathscr{H})$ depends on D, hence, differently from the proof of Lemma 3.3, we cannot apply Jensen's inequality and optimize t jointly over ξ and D.

Fix $S=(z_1,\dots,z_n)\in(\mathcal{X}\times\mathcal{Y})^n$ and let $V_S\coloneqq\{(\ell_h(z_1),\dots,\ell_h(z_n)):h\in\mathscr{H}\}$ be the set of loss vectors, so that $|V_S|=N_S(\mathscr{H})$. The quantity $\sum_i\xi_i\ell_h(z_i)$ depends

on h only through its loss vector, hence

$$\sup_{h\in\mathscr{H}} \sum_{i=1}^{n} \xi_i \ell_h(z_i) = \max_{v\in V_S} \sum_{i=1}^{n} \xi_i v_i.$$

Then, for any $t > 0$, proceeding as in the proof of Lemma 3.3 with the expectation over ξ only, we have

$$\exp\left(t\,\mathbb{E}_\xi\left[\max_{v\in V_S}\sum_{i=1}^{n}\xi_i v_i\right]\right) \le \mathbb{E}_\xi\left[\max_{v\in V_S}\exp\left(t\sum_{i=1}^{n}\xi_i v_i\right)\right] \le \sum_{v\in V_S}\prod_{i=1}^{n}\mathbb{E}_{\xi_i}\left[e^{t\xi_i v_i}\right]$$
$$\le \sum_{v\in V_S}\prod_{i=1}^{n} e^{t^2M^2/2} = N_S(\mathscr{H})\, e^{nt^2M^2/2},$$

where we use Jensen's inequality, we bound the maximum with the sum and use the independence of the ξ_i's, and we apply Hoeffding's lemma (Theorem A.5) to $\xi_i v_i$, which is centered and satisfies $|\xi_i v_i| = v_i \le M$. Taking the logarithm and dividing by tn,

$$\mathbb{E}_\xi\left[\sup_{h\in\mathscr{H}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(z_i)\right] \le \frac{\log N_S(\mathscr{H})}{tn} + \frac{tM^2}{2}.$$

Since S is fixed, we can choose t depending on S. Optimizing as in the proof of Lemma 3.3, with $t^\star = \sqrt{2\log N_S(\mathscr{H})/(nM^2)}$, gives

$$\mathbb{E}_\xi\left[\sup_{h\in\mathscr{H}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(z_i)\right] \le M\sqrt{\frac{2\log N_S(\mathscr{H})}{n}}$$

(if $N_S(\mathscr{H}) = 1$, the left-hand side is 0 since $\mathbb{E}[\xi_i] = 0$). Since this holds for every $S \in (\mathcal{X}\times\mathcal{Y})^n$, it holds for every realization of D, i.e. the inner expectation is bounded by $M\sqrt{2\log N_D(\mathscr{H})/n}$ for every realization of D. Taking the expectation over D yields the result. □

Proof of Lemma 3.5. We prove that $N_S(\mathscr{H}) \le |\mathscr{H}|$ and $N_S(\mathscr{H}) \le 2^n$ for every $S \in (\mathcal{X}\times\mathcal{Y})^n$. Fix $S = (z_1,\dots,z_n)$ and consider the map

$$\Phi\colon \mathscr{H} \to \{0,1\}^n, \qquad h \mapsto \big(\ell_h(z_1),\dots,\ell_h(z_n)\big),$$

so that $N_S(\mathscr{H}) = |\Phi(\mathscr{H})|$. Since the image of a set under any map has cardinality at most that of the domain, $|\Phi(\mathscr{H})| \le |\mathscr{H}|$. Since $\Phi(\mathscr{H}) \subseteq \{0,1\}^n$, we also have $|\Phi(\mathscr{H})| \le |\{0,1\}^n| = 2^n$. Because this holds for every realization S of D, the claim for $N_D(\mathscr{H})$ follows. □

Proof of Theorem 3.6. Since $\hat{h}_S \in \mathscr{H}$ for any realization S of D,

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \leq \mathbb{E}_D\left[\sup_{h\in\mathscr{H}}(R - \hat{R}_D)(h)\right]$$

The proof is concluded applying Theorem 3.2, bounding the Rademacher complexity with Lemma 3.4, and bounding $N_D(\mathscr{H})$ with Lemma 3.5. □

Proof of Lemma 4.1. Let $\mathscr{L} = \{(x, y) \mapsto \ell(y, h(x)) : h \in \mathscr{H}\}$, where ℓ is the zero-one loss. Apply the first inequality of Theorem 5.1 of [4], with $\mathbb{S}_{\mathcal{F}}(X_1^{2n}) = N_{2n}(\mathscr{H})$, yielding with probability at least $1 - \delta$, simultaneously for all $h \in \mathscr{H}$,

$$R(h) - \widehat{R}_D(h) \leq 2\sqrt{\frac{R(h)\log(4N_{2n}(\mathscr{H})/\delta)}{n}}.$$

Set

$$\delta_\varepsilon = 4N_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon < 1$, applying the preceding inequality with $\delta = \delta_\varepsilon$ gives

$$\Pr\left[\exists h \in \mathscr{H} : R(h) - \widehat{R}_D(h) > \varepsilon\sqrt{R(h)}\right] \leq 4N_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon \geq 1$, this probability bound holds trivially. Finally, since $n\varepsilon^2 \geq 8$,

$$4e^{-n\varepsilon^2/4} \leq 2e^{-n\varepsilon^2/16},$$

which proves the claim. □

Proof of Theorem 4.3. Let $\delta \in (0, 1)$ and set

$$u \coloneqq \frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n} \qquad \text{and} \qquad \varepsilon \coloneqq 4\sqrt{u},$$

so that $2N_{2n}(\mathscr{H})e^{-n\varepsilon^2/16} = \delta$. Since $N_{2n}(\mathscr{H}) \geq 1$ and $\delta < 1$, we have $n\varepsilon^2 = 16\log(2N_{2n}(\mathscr{H})/\delta) \geq 16\log 2 \geq 8$, hence Lemma 4.1 applies and gives

$$\Pr\left[\exists h \in \mathscr{H} : (R - \hat{R}_D)(h) > \varepsilon\sqrt{R(h)}\right] \leq \delta.$$

Therefore, with probability at least $1 - \delta$, we have $(R - \hat{R}_D)(h) \leq \varepsilon\sqrt{R(h)}$ for all $h \in \mathscr{H}$, and in particular for $h = \hat{h}_D \in \mathscr{H}$.

On this event, we isolate $\hat{R}_D(\hat{h}_D)$. Let $r \coloneqq R(\hat{h}_D)$ and $\hat{r} \coloneqq \hat{R}_D(\hat{h}_D)$, so that $r - \varepsilon\sqrt{r} - \hat{r} \leq 0$. This is a quadratic inequality in $x \coloneqq \sqrt{r} \geq 0$, namely $x^2 - \varepsilon x - \hat{r} \leq 0$, which implies that x is at most the largest root,

$$\sqrt{r} \leq \frac{\varepsilon + \sqrt{\varepsilon^2 + 4\hat{r}}}{2}.$$

Squaring both sides,

$$r \le \frac{2\varepsilon^2 + 4\hat{r} + 2\varepsilon\sqrt{\varepsilon^2 + 4\hat{r}}}{4} = \hat{r} + \frac{\varepsilon^2}{2} + \frac{\varepsilon}{2}\sqrt{\varepsilon^2 + 4\hat{r}} \le \hat{r} + \varepsilon^2 + \varepsilon\sqrt{\hat{r}},$$

where in the last inequality we used $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$. Hence, substituting $\varepsilon = 4\sqrt{u}$, with probability at least $1 - \delta$,

$$\begin{aligned}(R - \hat{R}_D)(\hat{h}_D) = r - \hat{r} &\le \varepsilon\sqrt{\hat{r}} + \varepsilon^2 \\ &= 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2N_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

Proof of Corollary 4.4. Recall $h_0 = \arg\min_{h \in \mathscr{H}} R(h)$, $R(h_0) = R^*_{\mathscr{H}}$ by definition of $R^*_{\mathscr{H}}$ and $\hat{R}_D(\hat{h}_D) \le \hat{R}_D(h_0)$ by definition of the ERM $\hat{h}_D$. Since h_0 is independent of the data, the variables $\ell(h_0(X_i), Y_i)$ are i.i.d. Bernoulli with mean $R^*_{\mathscr{H}}$ and variance $R^*_{\mathscr{H}}(1 - R^*_{\mathscr{H}}) \le R^*_{\mathscr{H}}$. Bernstein inequality (Theorem A.3) with $M = 1$ gives, with probability at least $1 - \delta/2$,

$$\hat{R}_D(h_0) \le R^*_{\mathscr{H}} + \sqrt{\frac{2R^*_{\mathscr{H}}\log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}.$$

Let $u \coloneqq \log(4N_{2n}(\mathscr{H})/\delta)/n$. Since $N_{2n}(\mathscr{H}) \ge 1$, $\log(2/\delta)/n \le u$. Thus

$$\hat{R}_D(\hat{h}_D) \le \hat{R}_D(h_0) \le R^*_{\mathscr{H}} + 2\sqrt{R^*_{\mathscr{H}} u} + u = \left(\sqrt{R^*_{\mathscr{H}}} + \sqrt{u}\right)^2. \tag{11}$$

Theorem 4.3 at confidence $1 - \delta/2$ gives, with probability at least $1 - \delta/2$,

$$(R - \hat{R}_D)(\hat{h}_D) \le 4\sqrt{\hat{R}_D(\hat{h}_D)\, u} + 16u. \tag{12}$$

By a union bound, both events hold simultaneously with probability at least $1 - \delta$. Hence, plugging (11) in the RHS of (12), we obtain that with probability at least $1 - \delta$,

$$\begin{aligned}(R - \hat{R}_D)(\hat{h}_D) &\le 4\sqrt{u}\left(\sqrt{R^*_{\mathscr{H}}} + \sqrt{u}\right) + 16u \\ &= 4\sqrt{\frac{R^*_{\mathscr{H}}\log(4N_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

Proof of Corollary 4.5. Substitute $R^*_{\mathscr{H}} = \sigma^2 + A_{\mathscr{H}}$ into Corollary 4.4. □

Proof of Theorem 4.6. Let $\mathrm{gap} = (R - \hat{R}_D)(\hat{h}_D)$ and $C = \sqrt{\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}}$. Corollary 4.4 gives

$$\Pr\left[\mathrm{gap} \le 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2\right] \ge 1 - \delta$$
$$\implies \Pr\left[\mathrm{gap} > 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2\right] \le \delta$$

Set $\varepsilon = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$. Let us now find $\delta(\varepsilon)$:

$$20C^2 + 4\sqrt{R^*_{\mathscr{H}}}\, C - \varepsilon = 0 \implies C = \frac{-4\sqrt{R^*_{\mathscr{H}}} + \sqrt{16R^*_{\mathscr{H}} + 80\varepsilon}}{40} = \frac{\sqrt{R^*_{\mathscr{H}} + 5\varepsilon} - \sqrt{R^*_{\mathscr{H}}}}{10}$$

So we have

$$\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n} = \left(\frac{\sqrt{R^*_{\mathscr{H}} + 5\varepsilon} - \sqrt{R^*_{\mathscr{H}}}}{10}\right)^2$$
$$\frac{4N_{2n}(\mathscr{H})}{\delta} = \exp\left(n\left(\frac{\sqrt{R^*_{\mathscr{H}} + 5\varepsilon} - \sqrt{R^*_{\mathscr{H}}}}{10}\right)^2\right)$$

and hence

$$\delta(\varepsilon) = 4N_{2n}(\mathscr{H}) \exp\left(-\frac{n}{100}\left(\sqrt{R^*_{\mathscr{H}} + 5\varepsilon} - \sqrt{R^*_{\mathscr{H}}}\right)^2\right).$$

Using that $\mathbb{E}[X] = \int_0^\infty \Pr[X \ge \varepsilon]\, d\varepsilon$ for $X \ge 0$, applied to $\mathrm{gap}^+ \coloneqq \max\{\mathrm{gap}, 0\}$,

$$\mathbb{E}_D[\mathrm{gap}] \le \mathbb{E}_D[\mathrm{gap}^+] = \int_0^\infty \Pr[\mathrm{gap} \ge \varepsilon]\, d\varepsilon \le \int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon.$$

We now compute the integral by exchanging the roles of ε and δ. The map $\varepsilon \mapsto \delta(\varepsilon)$ is continuous and strictly decreasing on $[0, \infty)$, with $\delta(0) = 4N_{2n}(\mathscr{H}) > 1$ and $\delta(\varepsilon) \to 0$ as $\varepsilon \to \infty$. Hence, for every $t \in (0, 1)$ there is a unique $\varepsilon(t) > 0$ such that $\delta(\varepsilon(t)) = t$, and it is given by $\varepsilon(t) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ with $C = \sqrt{\log(4N_{2n}(\mathscr{H})/t)/n}$, by construction of $\delta(\varepsilon)$. Moreover, for $t \in (0, 1)$,

$$\min\{1, \delta(\varepsilon)\} > t \iff \delta(\varepsilon) > t \iff \varepsilon < \varepsilon(t).$$

Writing $\min\{1, \delta(\varepsilon)\} = \int_0^1 \mathbf{1}\{t < \min\{1, \delta(\varepsilon)\}\}\, dt$ and exchanging the order of integration by Tonelli's theorem (the integrand is non-negative), we obtain

$$\int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon = \int_0^1 \int_0^\infty \mathbf{1}\{\varepsilon < \varepsilon(t)\}\, d\varepsilon\, dt = \int_0^1 \varepsilon(t)\, dt = \int_0^1 \varepsilon(\delta)\, d\delta.$$

Hence, writing $\varepsilon(\delta) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ explicitly,

$$\mathbb{E}_D[\mathrm{gap}] \le \int_0^1 \varepsilon(\delta)\, d\delta = 4\sqrt{\frac{R^*_{\mathscr{H}}}{n}} \int_0^1 \sqrt{\log \frac{4N_{2n}(\mathscr{H})}{\delta}}\, d\delta + \frac{20}{n} \int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta.$$

The logarithm is the only part depending on δ, and its integral is

$$\int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta = \log(4N_{2n}(\mathscr{H})) - \int_0^1 \log \delta \, d\delta = \log(4N_{2n}(\mathscr{H})) + 1.$$

Applying Jensen's inequality to the square root term,

$$\int_0^1 \sqrt{\log \frac{4N_{2n}(\mathscr{H})}{\delta}}\, d\delta \le \sqrt{\int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta} = \sqrt{\log(4N_{2n}(\mathscr{H})) + 1},$$

we obtain

$$\mathbb{E}_D[\text{gap}] \le 4\sqrt{\frac{R^*_{\mathscr{H}}\left(\log(4N_{2n}(\mathscr{H})) + 1\right)}{n}} + 20\,\frac{\log(4N_{2n}(\mathscr{H})) + 1}{n}.$$

Asymptotically,

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\,\log N_{2n}(\mathscr{H})}{n}} + \frac{\log N_{2n}(\mathscr{H})}{n}\right)$$

because, at the numerator,

$$\log(4N_{2n}(\mathscr{H})) + 1 = \log 4 + \log N_{2n}(\mathscr{H}) + 1 = \log N_{2n}(\mathscr{H}) + O(1)$$

and for large x,

$$\log x + O(1) \le C \log x$$

for some constant $C > 0$. Apply Lemma 3.5, obtaining

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le \min\left\{O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\log|\mathscr{H}|}{n}} + \frac{\log|\mathscr{H}|}{n}\right), O\left(\sqrt{A_{\mathscr{H}} + \sigma^2} + 1\right)\right\}$$

□

Proof of Lemma 5.2. We prove the inequalities for $N_S(\mathscr{H})$, for every $S \in (\mathcal{X}\times\mathcal{Y})^n$. Since $N_D(\mathscr{H})$ takes the value $N_S(\mathscr{H})$ when D takes the value S, they hold for $N_D(\mathscr{H})$ for every realization of D.

Fix $S = ((x_1, y_1), \dots, (x_n, y_n)) \in (\mathcal{X} \times \mathcal{Y})^n$, and consider the maps

$$\Psi\colon \mathscr{H} \to \mathcal{Y}^n, \quad h \mapsto \big(h(x_1), \dots, h(x_n)\big),$$

$$\lambda\colon \mathcal{Y}^n \to \{0,1\}^n, \quad (\hat{y}_1, \dots, \hat{y}_n) \mapsto \big(\ell(\hat{y}_1, y_1), \dots, \ell(\hat{y}_n, y_n)\big).$$

The map $\Phi\colon h \mapsto (\ell_h(x_1, y_1), \dots, \ell_h(x_n, y_n))$ of the proof of Lemma 3.5 factorizes as $\Phi = \lambda \circ \Psi$, hence $N_S(\mathscr{H}) = |\Phi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))|$. Moreover, by definition of growth function, $|\Psi(\mathscr{H})| \le \Pi_n(\mathscr{H})$.

$1 \le N_S(\mathscr{H})$. Since $\mathscr{H}$ is nonempty, $\Phi(\mathscr{H})$ is nonempty.

$N_S(\mathscr{H}) \leq \Pi_n(\mathscr{H})$. Since the image of a set under any map has cardinality at most that of the set,

$$N_S(\mathscr{H}) = |\lambda(\Psi(\mathscr{H}))| \leq |\Psi(\mathscr{H})| \leq \Pi_n(\mathscr{H}).$$

$\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$. Let $x_1, \ldots, x_n \in \mathcal{X}$ and let Ψ be the corresponding map. As above, $|\Psi(\mathscr{H})| \leq |\mathscr{H}|$, and $\Psi(\mathscr{H}) \subseteq \mathcal{Y}^n$ with $|\mathcal{Y}^n| = 2^n$, since $\mathcal{Y} = \{-1, 1\}$. Taking the supremum over $x_1, \ldots, x_n$ gives $\Pi_n(\mathscr{H}) \leq |\mathscr{H}|$ and $\Pi_n(\mathscr{H}) \leq 2^n$.

$\Pi_n(\mathscr{H}) = N_n(\mathscr{H})$. Taking the supremum over S in $N_S(\mathscr{H}) \leq \Pi_n(\mathscr{H})$ gives $N_n(\mathscr{H}) \leq \Pi_n(\mathscr{H})$. For the converse, let $x_1, \ldots, x_n \in \mathcal{X}$, take any $y_1, \ldots, y_n \in \mathcal{Y}$, and let $S = ((x_1, y_1), \ldots, (x_n, y_n))$. Since $\mathcal{Y} = \{-1, 1\}$ and ℓ is the zero-one loss, $\hat{y}_i = y_i$ if $\ell(\hat{y}_i, y_i) = 0$ and $\hat{y}_i = -y_i$ if $\ell(\hat{y}_i, y_i) = 1$. Hence λ is injective, and

$$|\Psi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))| = N_S(\mathscr{H}) \leq N_n(\mathscr{H}).$$

Taking the supremum over $x_1, \ldots, x_n$ gives $\Pi_n(\mathscr{H}) \leq N_n(\mathscr{H})$. □

Appendix B. Preliminary experiments

B.1. **Noise structure.** We generate the labels with a centered parabola (see Section B.2), so that roughly half of them are 1 and half -1 in the interval $[0, 10]$. Then, we inspect what happens if we use a Bernoullian noise instead of a Gaussian one, i.e. when data is generated as

$$y \leftarrow sign(h^*(x))b$$

for $b \sim \text{Bernoulli}(p)$. Notice that this noise flips the labels with a probability of $1 - p$. While from the first experiments it seemed that the algorithm achieved a higher test error, we postulate that this may be due to the portion of misclassified inputs, rather than to the noise structure itself. Hence, fixing $p = 0.8$, we look for the value of σ that achieves roughly the same portion of mislcassified inputs (20%). Figure 8 shows that this happens for $\sigma = 5.5$, and Figures 9 and 10 show that the two noise structures then yield essentially the same train and test errors.

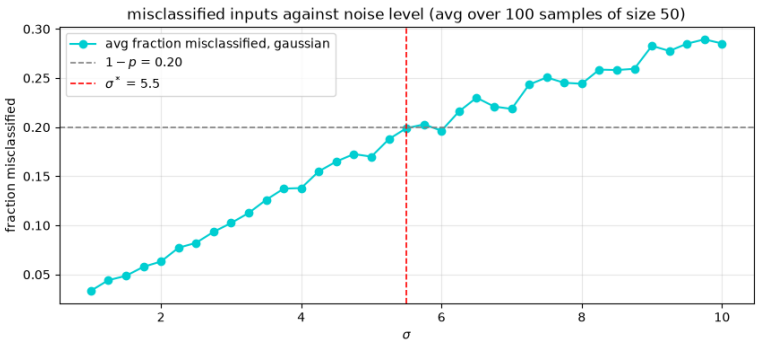


Figure 8. Calibrating Gaussian noise on the parabola. Average fraction of misclassified inputs against σ, over 100 samples of size $n = 50$.

B.2. **Data-generating function.** Our first choice of h^* is a parabola centered horizontally at 5, the midpoint of the support $\mathcal{X}$, and vertically so that roughly half of the points fall above the x-axis and half below. I.e.,

$$h^*(x) = (x - 5)^2 - 6.25$$

Figure 11 shows the data generated with the centered parabola, for different amounts of noise.

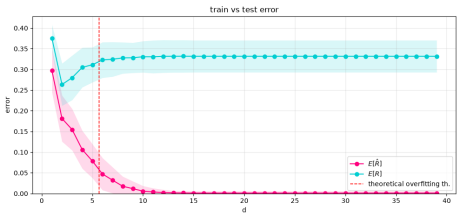


FIGURE 9. Parabola, Bernoulli noise with $p = 0.8$; $n = 50$, $N = 10,000$, $n_{\text{test}} = 200,000$.

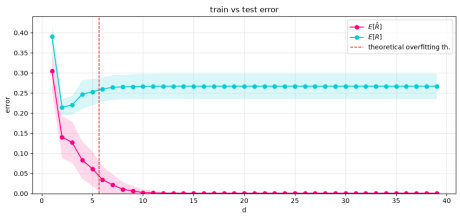


FIGURE 10. Parabola, Gaussian noise with $\sigma = 5.5$; $n = 50$, $N = 10,000$, $n_{\text{test}} = 200,000$.

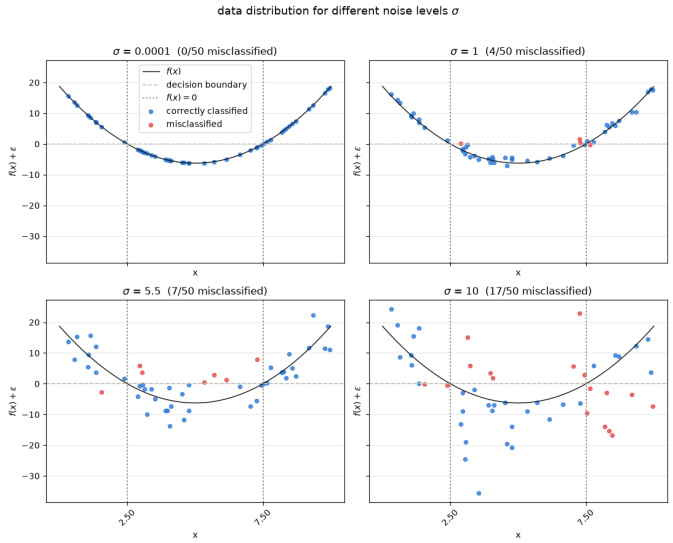


FIGURE 11. Samples from the parabola. One sample of size $n = 50$ for $\sigma \in \{10^{-4}, 1, 5.5, 10\}$.

Because in theory an algorithm could learn such a classification pattern as soon as it sees 3 points (one point for each region), we investigate what happens when the underlying function is more complex. We study a sinusoid with varying frequency,

$$h^*(x) = 5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right)$$

We multiply the sine by 5 to allow for enough height to properly inspect the data, while the argument of the sine is such that m is the number of times h^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of $1s$ and $-1s$.

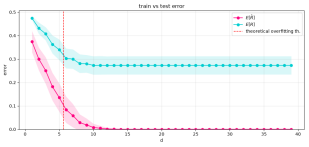


FIGURE 12. Sine, $m = 10$.

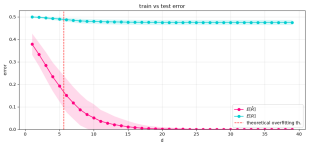


FIGURE 13. Sine, $m = 100$.

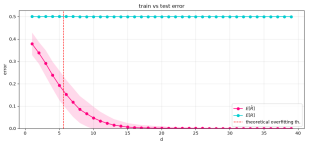


FIGURE 14. Sine, $m = 1000$.

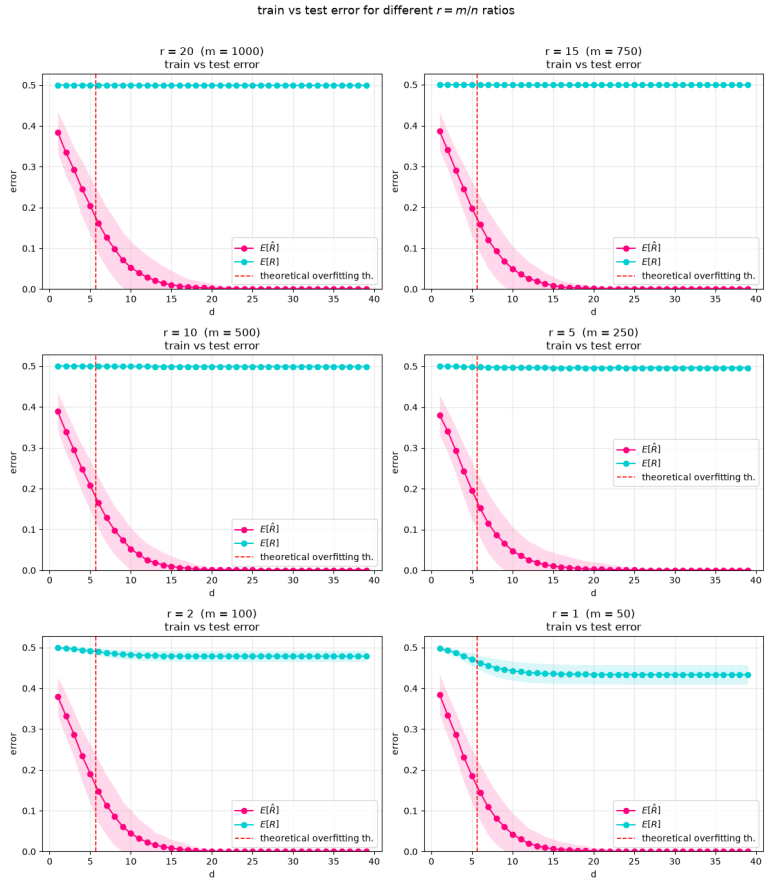


FIGURE 15. Learnability threshold in m/n. Train and test error for different ratios $r = m/n$; $n = 50$, $\sigma = 3$, $N = 10,000$, $n_{\text{test}} = 200,000$.

We explore how a random sample and the performance of the algorithm change for different sine frequencies. In particular, we expect that for a high frequency

the algorithm will not learn, as our datasize is $n = 50$. In Figures 12–14, we test a function that osciallets $10, 100$, and 1000 times with a noise level of $\sigma = 3$ (we reduce it compared to $\sigma = 5$, having increased the complexity of the function). We notice that for $m = 1000$ the algorithm fails to learn, and the test error stays stable at 0.5. This is because the labels change so frequently with respect to the number of points in the training dataset n that it is as if we had random labels. Hence, we investigate the relation between the complexity of the underlying function m and the data size n. We find that for $\frac{m}{n} > 5$, roughly, the algorithm ceases to learn (Figure 15). In the following experiments, we fix $m = 10$ and $n = 50$, which is very far from the threshold.

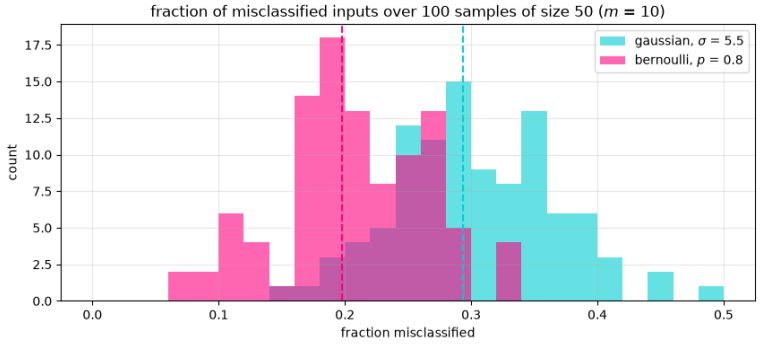


FIGURE 16. Gaussian against Bernoulli noise on the sine ($m = 10$). Fraction of misclassified inputs over 100 samples of size $n = 50$; dashed lines are the averages.

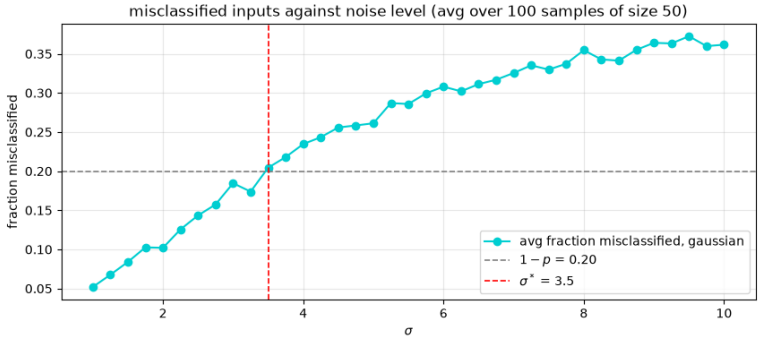


FIGURE 17. Calibrating Gaussian noise on the sine ($m = 10$). Average fraction of misclassified inputs against σ, over 100 samples of size $n = 50$.

We then verify whether a Gaussian noise with $\sigma = 5.5$ achieves the same portion of misclassified inputs on average as a Bernoullian noise with $p = 0.8$, as was the case for data generated with a parabola. Figure 16 shows that this is not the case on average (29% against 20%), while Figure 17 shows that the desired value is $\sigma = 3.5$.

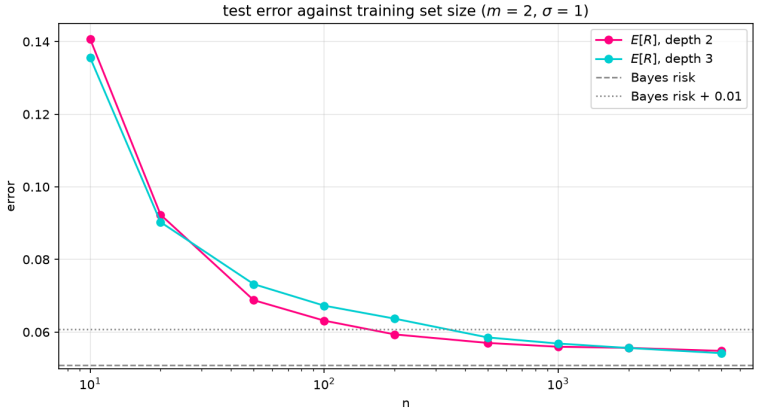


FIGURE 18. Sample size needed for a simple function, $\sigma = 1$. Test error at depth 2 and 3 against n for the sine with $m = 2$, averaged over 100 samples; $N = 10,000$, $n_{\text{test}} = 200,000$.

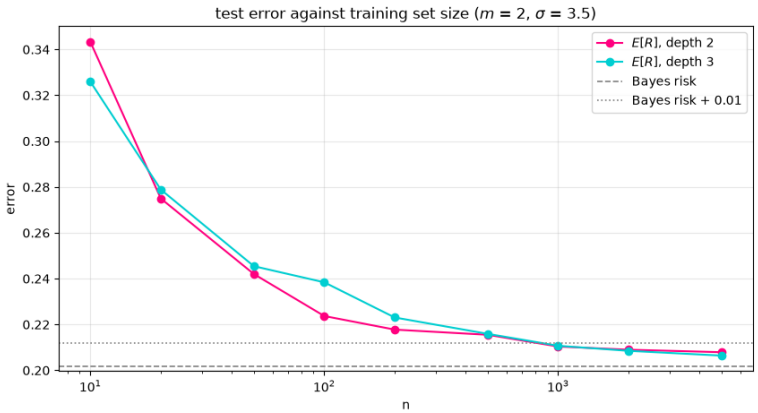


FIGURE 19. Sample size needed for a simple function, $\sigma = 3.5$. Test error at depth 2 and 3 against n for the sine with $m = 2$, averaged over 100 samples; $N = 10,000$, $n_{\text{test}} = 200,000$.

Finally, we investigate the afore-mentioned intuition about how easy it is for the algorithm to learn a simple function. We take $m = 2$ (which corresponds to a

parabola, in terms of classification pattern), sample uniformly from $\mathcal{X}$, and look for the training set size n for which the test error gets within 0.01 of the Bayes risk on average, at depth 2 and 3.

We start with little noise, $\sigma = 1$, for which the Bayes risk is ≈ 0.05 (Figure 18). Here the algorithm learns the function quickly: with $n = 50$ points the test error is already ≈ 0.07, within 0.02 of the Bayes risk, and it gets within 0.01 with $n = 200$ at depth 2 and $n = 500$ at depth 3.

With our noise level, $\sigma = 3.5$, the Bayes risk is ≈ 0.20 and $n = 1000$ training points are needed on average for the test error to get within 0.01 of it, both at depth 2 and 3 (Figure 19). Hence, although two splits are enough to represent the function, with this amount of noise the algorithm needs many points to place them correctly.

Appendix C. Additional results

Fact C.1 (The empirical risk is a biased estimate of the true risk for the ERM)**.** *lalala*

Fact C.2 (Bound on the Rademacher complexity of $\mathscr{L}$)**.** *lalalala*